MEHMET KEREM TURKCAN

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SUMMARY

Associate Research Scientist at Columbia University with 8+ years of experience building computer vision, multimodal, and scientific data systems for environments where latency, reliability, and real-world constraints matter. I lead research from data and model design through optimization, deployment, evaluation, and public release, with work spanning urban infrastructure, medical robotics, and connectome-scale neuroscience. My record includes systems deployed across major urban testbeds, publications at CVPR, ICML, ACM UIST, ACM/IEEE SEC, IEEE INFOCOM, IJCV, and eLife, and open-source projects with 1,200+ combined GitHub stars.

SELECTED CONTRIBUTIONS

- ◇ **Real-world AI systems:** Lead multimodal perception and urban analytics efforts intended to scale across 900+ New York City intersections and the COSMOS PAWR, DataCity, and FAU MobIntel testbeds.
- ◇ **Frontier model deployment:** Created DART, a real-time open-vocabulary detector with TensorRT deployment, public weights, and 300+ GitHub stars; led UrbanOmniDetect, an oral paper at the CVPR 2026 DriveX Workshop.
- ◇ **Sponsored research:** Contributed proposal development, technical reporting, sponsor reviews, and engineering delivery across approximately $30M in institutional research supported by NSF, DARPA, AFOSR, and Con Edison, with additional project support from NVIDIA and EmpireAI.
- ◇ **Open scientific platforms:** Built FlyBrainLab in TypeScript with Jupyter Lab integration, OrientDB-backed graph and ontology systems, large-scale query layer, retrieval and question answering workflows, GPU simulation, and interactive 3D visualization.
- ◇ **Teaching and mentorship:** Taught Columbia graduate deep learning courses, mentored Master's and high school researchers, and served as the main engineering instructor for CS3's Research Experience for Teachers in 2024–2026.

EXPERIENCE

◇ **Columbia University**, New York, NY — Jul 2024 – Present

Associate Research Scientist, Civil Engineering & Engineering Mechanics

- Lead large-scale applied AI projects in real-time video analytics, open-vocabulary and 3D perception, edge/VLM systems, urban digital twins, and medical robotics foundation models; manage shared high-performance computing and cross-team engineering.
- Created DART, converting promptable frontier vision models into a real-time multiclass detector through shared backbone inference, batched decoding, adapter distillation, and TensorRT FP16 optimization.
- Led UrbanOmniDetect, a calibration-free, view-agnostic monocular 3D detector for ego-vehicle, infrastructure, and aerial cameras, with public models and the UrbanOmniView dataset.
- Contributed to Open-H-Embodiment and surgical world model research for healthcare robotics; developed public model assets for video generation, phase recognition, metric depth, and skills assessment.
- Co-authored the ACM/IEEE SEC 2025 Best Paper on privacy-preserving urban safety analytics, ICML 2025 work on adaptive data collection, and 2026 papers at CVPR Workshops and IEEE INFOCOM.
- Designed and led the engineering curriculum for three annual CS3 Research Experience for Teachers cohorts, translating current AI research into reproducible K–12 lesson plans and classroom projects.

◇ **Columbia University**, New York, NY — Nov 2022 – Jul 2024

Postdoctoral Research Scientist

- Built real-time digital twins of New York City intersections integrating object detection, multi-object tracking, trajectory forecasting, simulation, and safety warning generation.
- Developed synthetic data and benchmarking systems for dense urban perception, including Boundless in Unreal Engine 5 and the Constellation high-altitude object detection dataset.
- Built healthcare AI pipelines with Northwell Health collaborators for surgical phase recognition and automated endoscopy skill scoring, resulting in peer-reviewed journal publications and clinical presentations.
- Mentored 11 Master's researchers and three high school researchers across computer vision, generative modeling, and urban analytics projects.

◇ **Columbia University**, New York, NY — Jan 2023 – Dec 2023

Lecturer in Electrical Engineering

- Designed and taught ECBM E4040 Neural Networks & Deep Learning and EECS E6691 Advanced Deep Learning, including lectures, coding assignments, exams, and conference-style final projects.
- Co-taught VOAI 0003E Neural Networks and Deep Learning for the Columbia Video Network AI Executive Certificate in Fall 2022 and Summer 2023.

◇ **Columbia University**, New York, NY Jan 2017 – Oct 2022
Doctoral Researcher, Bionet Group

- Co-created FlyBrainLab, an open computing platform for natural language exploration, graph querying, simulation, and visualization of connectome-scale neuroscience data.
- Developed an early research agent architecture combining programmable ontologies, literature-linked entity retrieval, dense passage retrieval, biomedical BERT question answering, graph database tools, and executable scientific workflows.
- Built scalable TypeScript and Python interfaces, OrientDB/NeuroArch query services, GPU-backed neural simulations, and interactive 3D tools used in open neuroscience research and education.

◇ **Columbia University and ArtGe Technologies** 2014 – 2016
Research and R&D Roles New York, NY and Istanbul

- Developed deep learning systems for thermal camera infrastructure inspection, solar microgrid control, intelligent building forecasting, stadium-scale face recognition, and agricultural stereo vision.

EDUCATION

◇ **Columbia University**, New York, NY 2022
- Ph.D. in Electrical Engineering; Systems Biology and Neuroengineering, GPA 4.10.

◇ **Columbia University**, New York, NY 2016
- M.S. in Computer Science; Machine Learning and Thesis Track, GPA 3.97.

◇ **Istanbul Technical University**, Istanbul, Turkey 2015
- B.S. in Electronics and Communication Engineering, GPA 3.75.

AWARDS AND RECOGNITION

◇ **Best Paper Award**, ACM/IEEE Symposium on Edge Computing 2025
◇ NVIDIA research grant, *Edge AI for Equitable and Safe Intersections in Urban Metropolises* 2025
◇ EmpireAI award, *AI Computing for Community, Health, Safety and Climate* 2025
◇ INRIX x MetroLab Challenge finalist 2025
◇ Columbia University Electrical Engineering Distinguished Teaching Award 2024
◇ Smart Cities North America Award winner in Transportation, IDC Government Insights 2023
◇ National Postdoc Appreciation Week Excellence Award, Columbia University 2023
◇ Herbert French Fellowship, Columbia University; Helmsley Fellowship, CSHL 2017–2019

SPONSORED RESEARCH CONTRIBUTIONS

Institutional award totals; dates indicate my period of contribution.

◇ **NSF Engineering Research Center for Smart Streetscapes (CS3)** (EEC-2133516) 2022 – Present
Research staff; research coordination, annual reporting, and sponsor reviews $26M center award

- Coordinate applied AI research and cross-team demonstrations across a five-institution center; present DART, UrbanOmniDetect, and related systems at annual NSF site visits.

◇ **NSF CPS: Hybrid Twins for Urban Transportation** (CNS-2038984) 2021 – Present
Technical, reporting, and sponsor review lead $1.3M Columbia award

- Led engineering delivery, prepared annual technical reports, and presented project reviews to NSF for traffic sensing, prediction, and digital twin systems spanning individual intersections and citywide operations.

◇ **DARPA Robust Learning in Brain Circuits of Synthetic Miniature Insects** (HR0011-19-9-0035) 2019 – 2020
Proposal and technical report contributor $620K Columbia award

- Proposal and technical report contributor; contributed to proposal development, authored project reports, and conducted grant-supported doctoral research on robust learning and executable brain circuit models.

◇ **AFOSR Foundations of Neuroinformation Processing** (FA9550-16-1-0410) 2017 – 2021
Grant-supported doctoral researcher $1.11M Columbia award

- Developed computational systems and models for phase- and spike-based neural information processing, including work underlying the FlyBrainLab platform.

◇ **NSF Collaborative Research: The Digital Fly Brain** (DBI-1544383) 2017 – 2019
Grant-supported doctoral researcher $795K Columbia award

- Engineered connectome data, graph query, simulation, and visualization capabilities for the open-source Fruit Fly Brain Observatory and FlyBrainLab systems.

◇ **Con Edison Underground Infrastructure and Public Safety Analytics** (CU09-1331) 2015 – 2016
Graduate Researcher; Amendments 21–23 Con Edison

- Developed theoretical models, implementations, and data processing pipelines for a Visual Data Capture pilot using thermal imagery, high-resolution photographs, and video of underground electrical structures; examined automated hotspot detection, inspection targeting, and estimation of avoided serious events.
- Extended the system with infrared camera data assimilation, equipment and cable density models, temperature differential extraction along cable lengths, and visual features associated with later failures; delivered project outputs to Con Edison.
- Contributed cost-benefit and portfolio optimization analysis for secondary public safety programs.

SELECTED SOFTWARE AND DATA

◇ **DART:** Real-time open-vocabulary detection; PyTorch, TensorRT, distillation, public weights, and deployment benchmarks.
◇ **UrbanOmniDetect:** Calibration-free monocular 3D detection with public models and the UrbanOmniView multi-view dataset.
◇ **Loom:** Analytical looped-transformer computer with C compilation, ONNX/WebGPU demos, FPGA verification, and 140+ tests.
◇ **FlyBrainLab:** Natural language and ontology interfaces, OrientDB graph querying, GPU simulation, and 3D scientific visualization.
◇ **Boundless:** Unreal Engine 5 synthetic data pipeline and datasets for infrastructure and aerial object detection.
◇ **bikeped:** Real-time fisheye perception, safety alerts, simulator scenarios, and conformance testing for urban intersections.
◇ **GPTune:** Early GPT-2 fine-tuning and sampling toolkit for custom language model adaptation.

SELECTED PUBLICATIONS AND PREPRINTS

1. **Constellation Dataset: Benchmarking High-Altitude Object Detection for an Urban Intersection. Mehmet Kerem Turkcan**, C. Zang, S. Narasimhan, G. H. Je, B. Yu, M. Ghasemi, J. Ghaderi, G. Zussman, and Z. Kostic. *International Journal of Computer Vision, 134(10):429, 2026.*
2. **Calibration-Free View-Agnostic Monocular 3D Object Detection for Urban Scenes. Mehmet Kerem Turkcan**, D. Gumaste, and Z. Kostic. *CVPR Workshops (DriveX), oral presentation, 2026.*
3. **AI-Powered CPS-Enabled Vulnerable-User-Aware Urban Transportation Digital Twin: Methods and Applications.** Y. Fu, **Mehmet Kerem Turkcan**, M. Ghasemi, Z. Mo, C. Zang, A. Adhikari, Z. Kostic, G. Zussman, and X. Di. *IEEE Transactions on Intelligent Transportation Systems, 27(5):4949–4966, 2026.*
4. **Worst-Case Attacks in Reactive Edge-Cloud Systems: Expected Latency and SLA Violation.** J. Tavori, **Mehmet Kerem Turkcan**, Z. Kostic, J. Ghaderi, and G. Zussman. *IEEE INFOCOM, 2026.*
5. **Digital Eyes on the Road: Using Street Cameras to Verify Traffic Integrity and Detect Sybil Attacks at City Scale.** J. Tavori, **Mehmet Kerem Turkcan**, Z. Kostic, S. Stolfo, and G. Zussman. *USENIX VehicleSec, 2026.*
6. **Harnessing Floating Car Data, Traffic Camera Observations, and Network Flow Analysis for Traffic Volume Estimation.** A. Kosikova, **Mehmet Kerem Turkcan**, A. Darrat, and A. Smyth. *Discover Civil Engineering, 3(1):257, 2026.*
7. **Newell-Car-Following-Model-Informed Gaussian Process Regression Model for Ultra-Short-Term Arterial Vehicle Trajectory Prediction.** N. S. Ahmad, P. J. Jin, C. Zang, **Mehmet Kerem Turkcan**, and Z. Kostic. *Transportation Research Record, to appear, 2026.*
8. **Zero-Shot Image Tagging at Traffic Intersections via SAM-Guided Vision-Language Models.** W. Ho, **Mehmet Kerem Turkcan**, J. Ghaderi, G. Zussman, and Z. Kostic. *ICMLA, 2026.*
9. **Computer Vision Scoring of Endoscopically Traced Line Figures in an Inanimate Scope Tip Coordination Training Model.** N. Mitra, S. Narasimhan, **Mehmet Kerem Turkcan**, et al. *Surgical Endoscopy, 40(1):657–664, 2026.*
10. **Dynamic Minimax Regret Optimization for Robust LLM Post-Training.** C. Zang, H. Dong, **Mehmet Kerem Turkcan**, G. Zussman, Z. Kostic, and J. Ghaderi. *arXiv:2610.06329, 2026.*
11. **Surgical Kinematics from Monocular Video with Learned Articulated Motion Constraints. Mehmet Kerem Turkcan**, S. Samal, and Z. Kostic. *arXiv:2609.27227, 2026.*
12. **Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics.** N. Nelson, J.-T. Chen, J. Haworth, et al., including **Mehmet Kerem Turkcan**. *arXiv:2604.21017, 2026.*
13. **A Vision-Based Analysis of Congestion Pricing in New York City. Mehmet Kerem Turkcan**, J. Tavori, J. Ghaderi, G. Zussman, Z. Kostic, and A. Smyth. *arXiv:2602.03015, 2026.*
14. **Real-Time Video Analytics for Urban Safety: Deployment over Edge and End Devices.** M. Ghasemi, Y. Fu, X. Ouyang, P. Wang, **Mehmet Kerem Turkcan**, et al. *ACM/IEEE SEC, Best Paper Award, 2025.*
15. **Adaptive Data Collection for Robust Learning Across Multiple Distributions.** C. Zang, **Mehmet Kerem Turkcan**, G. Zussman, Z. Kostic, and J. Ghaderi. *ICML, 2025.*
16. **Distributed VLMs: Efficient Vision-Language Processing through Cloud-Edge Collaboration.** Y. Li, D. Gumaste, **Mehmet Kerem Turkcan**, J. Ghaderi, G. Zussman, and Z. Kostic. *IEEE PerCom Workshops, 2025.*

17. **Towards Suturing World Models: Learning Predictive Models for Robotic Surgical Tasks. Mehmet Kerem Turkcan**, M. Ballo, F. Filicori, and Z. Kostic. *arXiv:2503.12531, 2025.*
18. **StreetNav: Leveraging Street Cameras to Support Precise Outdoor Navigation for Blind Pedestrians.** G. Jain, B. Hindi, Z. Zhang, et al., including **Mehmet Kerem Turkcan**. *ACM UIST, 2024.*
19. **Boundless: Generating Photorealistic Synthetic Data for Object Detection in Urban Streetscapes. Mehmet Kerem Turkcan**, Y. Li, C. Zang, J. Ghaderi, G. Zussman, and Z. Kostic. *arXiv:2409.03022, 2024.*
20. **A Programmable Ontology Encompassing the Functional Logic of the Drosophila Brain. Mehmet Kerem Turkcan**, A. A. Lazar, and Y. Zhou. *Frontiers in Neuroinformatics, 2022.*
21. **Accelerating with FlyBrainLab the Discovery of the Functional Logic of the Drosophila Brain. Mehmet Kerem Turkcan**, A. A. Lazar, T. Liu, and Y. Zhou. *eLife, 2021.*

TEACHING AND CURRICULUM LEADERSHIP

◇ **Main Engineering Instructor, CS3 Research Experience for Teachers** 2024 – 2026
Center for Smart Streetscapes New York, NY
- Designed and led a four-week curriculum covering neural networks, model evaluation, computer vision, object tracking, generative AI, annotation, YOLO training, and edge deployment; supported teachers in producing classroom-ready lesson plans.

◇ **Lecturer, Columbia University** 2022 – 2023
Advanced Deep Learning; Neural Networks and Deep Learning New York, NY
- Taught EECS E6691, ECBM E4040, and VOAI 0003E across Fall 2022, Spring 2023, Summer 2023, and Fall 2023.

◇ **Course and Teaching Support, Columbia University** 2017 – 2021
Computational neuroscience and neuroengineering courses New York, NY
- Supported Computing with Brain Circuits of Model Organisms, Fruit Fly Brain as a Neuroinformation Processor, and Computational Neuroscience.

SELECTED PRESENTATIONS

◇ **CS3 NSF Annual Site Visit** May 12–15, 2026
DART and UrbanOmniDetect research demonstrations Columbia University
- Presented multiple live applied AI systems during the center's annual NSF review.

◇ **From Sensors to Systems: Real-Time Traffic Analysis for Faster Decision-Making** Feb. 5, 2026
Panelist, CS3 Innovation Summit Columbia University
- Discussed real-time sensing, edge AI, and operational urban analytics for faster public-sector decision-making.

◇ **Port Authority Tomorrow Summit** Jan. 7, 2026
Workshop host World Trade Center, New York
- Led a technology workshop at the Port Authority of New York and New Jersey headquarters.

◇ **Evaluating Micromobility & Dangerous Riding Behaviors** Nov. 19, 2025
Eighth Annual Vision Zero Research on the Road New York, NY
- Presented camera-based methods for evaluating micromobility patterns and dangerous riding behavior to New York City transportation stakeholders.

◇ **FlyBrainLab: An Interactive Open Computing Platform** Nov. 2020
PyData Global Online
- Presented the platform's graph query, neural simulation, and interactive scientific computing architecture.

PROFESSIONAL SERVICE

Peer reviewer: CVPR 2026; ICML 2026; ICMLA 2026; WACV 2025; ACM MobiCom 2024; IEEE/ACM ToN; IEEE TVLSI; Sensors, Electronics, and Smart Cities (MDPI).

TECHNICAL SKILLS

◇ **AI and deployment:** PyTorch, JAX, TensorFlow, Hugging Face, Ultralytics, OpenCV, TensorRT, ONNX, LiteRT, NCNN, WebGPU; NVIDIA Jetson, Qualcomm, and Raspberry Pi edge devices; detection, segmentation, tracking, 3D perception, VLM/VLA systems, LLM post-training (DPO, RL), RAG, agents, diffusion, and world models.
◇ **Simulation and robotics:** NVIDIA Isaac Lab and Isaac Sim, Newton (MuJoCo-Warp), Unreal Engine 5 (UE5), CARLA, OpenUSD, Blender; humanoid robot learning for the Unitree G1 (PPO, teacher-student distillation); city-scale digital twins, traffic and pedestrian simulation, synthetic data, and hardware-in-the-loop testing.
◇ **Languages, systems, and data:** Python, C/C++, CUDA, TypeScript/JavaScript, Node.js, SQL, MATLAB; OrientDB, RPC, distributed querying, HPC, Docker, GitHub Actions; large-scale data collection and annotation (CVAT), OpenStreetMap and NYC Open Data, JupyterLab, Three.js/WebGL, scientific visualization, and Adobe Creative Suite.